# Ch. 10 Non-Textbook Questions

Principles of Microeconomics

November 3, 2026

1. What type of externality (positive or negative) is present in each of the following examples? Is the social value of the activity greater than or equal to the private value? Is the social cost of the activity greater than or equal to the private cost? Without intervention, will there be too little or too much (relative to what would be efficient) of this activity?

   (a) Mr. Chau plants lots of colorful flowers in his front yard.

   (b) Your next-door neighbor likes to build bonfires in his backyard, and sparks often drift onto your house.

   (c) Maija, who lives next to an apple orchard, decides to keep bees to produce honey.

   (d) Justine buys a large SUV that consumes a lot of gasoline.

2. Suppose that tanneries pollute nearby bodies of water. The supply and demand for leather goods are given by $Q_S = \frac{1}{2}P$ and $Q_D = 10 - 2P$. If the pollution creates a \$3 per unit externality, please answer the following.

   (a) What are the price and quantity at the market equilibrium?

   (b) What are the price and quantity at the efficient equilibrium?

   (c) What is deadweight loss at the market equilibrium?

   (d) What is total surplus at the efficient equilibrium?

3. Supply for solar cells is given by $Q_S = 4P$, and demand is given by $Q_D = 14 - 2P$. Because they reduce carbon emissions, every solar cell sold generates \$2 worth of external value.

   (a) What quantity of solar cells is sold at the market equilibrium?

   (b) What quantity of solar cells is sold at the efficient equilibrium?

   (c) What government policy would generate the efficient equilibrium?

4. Suppose the government implemented a carbon tax of \$20 per ton of carbon dioxide emissions.

   (a) Draw a supply-and-demand diagram for carbon dioxide emissions. Indicate on your diagram the equilibrium price and quantity of emissions with and without the tax.

   (b) Demonstrate on your diagram what would happen to the price and quantity of emissions if the government raised the carbon tax.

   (c) Suppose the government replaced the carbon tax with a cap-and-trade system, but they targeted the same quantity of pollution as they did with the \$20 tax. Draw a new supply-and-demand diagram, and indicate the equilibrium price and quantity of emissions under the cap-and-trade system.

   (d) If one permit allows for one ton of carbon dioxide emissions, what is the price of a pollution permit under the cap-and-trade system?